
A Unifying Proximal-Splitting Template for the Convolutional Dictionary Update

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Abstract

Convolutional dictionary learning alternates a sparse-coding step with a dictionary-update step, and the latter is the expensive one: its frequency-domain normal equations factor per frequency into a diagonal-plus-rank- K system, where K is the number of training images. Every known update algorithm is a different way of handling this rank- K structure. We show that the differently-derived updates—an equality-constrained ADMM solved by iterated Sherman–Morrison or by conjugate gradient, a consensus ADMM, and an accelerated proximal-gradient (FISTA) update—are all instances of a single operator-splitting master problem parameterized by a small set of design axes. One shared driver instantiates each by axis choice and reproduces the reference solvers to near machine precision. The same template expresses the two multi-channel representations of the problem: a multi-channel dictionary with a shared coefficient map, and a single-channel dictionary with per-channel maps. Because the template is a genuine design space rather than a dispatcher, it predicts a method none of the known algorithms occupy—a *partial* consensus update whose blocks carry a rank- r Gram, interpolating between consensus ($r=1$) and the coupled solve ($r=K$)—and it re-derives, as a consequence of axis choices, the equivalence that the equality-constrained and consensus updates are the two endpoints of one block-partition axis, with the channel count entering that axis exactly as the image count does.

1 Introduction

A convolutional sparse representation models a signal $\mathbf{s}$ as a sum of a few filters convolved with sparse coefficient maps, $\mathbf{s} \approx \sum_m \mathbf{d}_m * \mathbf{x}_m$. Learning the filters is *convolutional dictionary learning* (CDL), which alternates a sparse-coding step (update the coefficient maps with the filters fixed) and a dictionary-update step (update the filters with the maps fixed). The dictionary-update step is the difficult one. With the coefficient maps held fixed, its normal equations in the DFT domain decompose per frequency into systems whose data-fidelity part is a diagonal matrix plus a rank- K term, where K is the number of training images. The sparse-coding step, in contrast, has a rank-1 structure per frequency, which is why the Sherman–Morrison shortcut makes sparse coding fast but does not transfer directly to the dictionary update [1].

A comparative study of convolutional dictionary learning [1] identified the dictionary-update step as the design bottleneck, compared the known update algorithms, and—most importantly for the present work—observed that two methods derived independently and presented as distinct are in fact the same class: the frequency-domain “3D” method is the consensus method with its data term and constraint expressed in the DFT basis. That observation is the seed of the present contribution. If different-looking updates are secretly instances of one structure, the structure can be written down as a template whose axes generate the known methods—and whose empty cells generate new ones.

We make three contributions. First, we state a single operator-splitting master problem for the dictionary update and identify the design axes that parameterize it; the four shipped update families are recovered by choosing axis values, and we validate this by matching a reference implementation

to near machine precision through one shared driver. Second, we express the two multi-channel representations of the problem as instances of the same template, and run both to convergence on color data. Third, we show the template predicts more than it was built from: a partial-consensus update that no known method occupies, and a re-derived equivalence—the equality-constrained and consensus updates are the two endpoints of one block-partition axis, and the channel count enters that axis exactly as the image count does. Throughout, the object of study is the dictionary update itself; the sparse-coding stage and the alternation are held fixed, and the frozen functional and constraint are those of the reviewed work [1].

2 The convolutional dictionary update

CDL minimizes, over filters $\{\mathbf{d}_m\}$ and coefficient maps $\{\mathbf{x}_{m,k}\}$,

$$\arg \min_{\{\mathbf{d}_m\}, \{\mathbf{x}_{m,k}\}} \frac{1}{2} \sum_k \left\| \sum_m \mathbf{d}_m * \mathbf{x}_{m,k} - \mathbf{s}_k \right\|_2^2 + \lambda \sum_{m,k} \|\mathbf{x}_{m,k}\|_1 \quad \text{s.t.} \quad \|\mathbf{d}_m\|_2 = 1 \quad \forall m, \quad (1)$$

over K training images. The unit-norm constraint removes the scaling ambiguity between filters and coefficients. The problem is non-convex jointly but convex in each block, and is solved by alternating a sparse-coding step with a dictionary-update step, coupled through their auxiliary variables with a single inner iteration of each before alternating [1].

With the coefficient maps fixed and re-indexed as $\mathbf{x}_{k,m}$, the dictionary update is

$$\arg \min_{\{\mathbf{d}_m\}} \frac{1}{2} \sum_k \left\| \sum_m \mathbf{x}_{k,m} * \mathbf{d}_m - \mathbf{s}_k \right\|_2^2 \quad \text{s.t.} \quad \mathbf{d}_m \in C_{\text{PN}}, \quad (2)$$

where C_{PN} is the support-plus-unit-norm constraint set (support restriction followed by per-filter normalization). Let $X_{k,m}$ be the convolution operator $X_{k,m} \mathbf{d}_m = \mathbf{x}_{k,m} * \mathbf{d}_m$, stack the filters as $\mathbf{d} = (\mathbf{d}_0^T, \mathbf{d}_1^T, \dots)^T$, and stack the signals as $\mathbf{s} = (\mathbf{s}_0^T, \mathbf{s}_1^T, \dots)^T$ with the block-convolution matrix $X = (X_{k,m})_{k,m}$. Then (2) is

$$\arg \min_{\mathbf{d}} \frac{1}{2} \|X\mathbf{d} - \mathbf{s}\|_2^2 + \iota_{C_{\text{PN}}}(\mathbf{d}), \quad (3)$$

with $\iota_{C_{\text{PN}}}$ the indicator of C_{PN} . In the DFT domain the associated normal equations $(\hat{X}^H \hat{X} + \sigma I) \hat{\mathbf{d}} = \hat{X}^H \hat{\mathbf{s}} + \sigma(\hat{\mathbf{g}} - \hat{\mathbf{h}})$ factor per frequency ω into N independent systems whose left-hand side is a diagonal matrix plus the rank- K term $\sum_k \hat{\mathbf{x}}_{k,\omega} \hat{\mathbf{x}}_{k,\omega}^H$. Every update algorithm is a way of handling this rank- K structure.

3 A unified operator-splitting template

The observation that organizes everything is that (3) has one data term $f(\mathbf{d}) = \frac{1}{2} \|X\mathbf{d} - \mathbf{s}\|_2^2$ that decomposes over *data blocks* (groupings of the image index) and one non-smooth term $\iota_{C_{\text{PN}}}$ whose proximal operator is cheap. We therefore write the data term as a sum over blocks b ,

$$f(\mathbf{d}) = \sum_{b=1}^B f_b(\mathbf{d}), \quad f_b(\mathbf{d}) = \frac{1}{2} \|X_b \mathbf{d} - \mathbf{s}_b\|_2^2, \quad (4)$$

and solve the *generalized consensus* master problem

$$\min_{\{\mathbf{d}_b\}, \mathbf{g}} \sum_{b=1}^B f_b(\mathbf{d}_b) + \iota_{C_{\text{PN}}}(\mathbf{g}) \quad \text{s.t.} \quad \mathbf{d}_b = \mathbf{g} \quad \forall b. \quad (5)$$

Each block b owns a copy $\mathbf{d}_b$ of the dictionary and a slice of the data; a global variable $\mathbf{g}$ carries the constraint. The design space lives in how (5) is partitioned, enforced, solved, represented, and transformed.

Design axes. The template is parameterized by five axes.

Table 1: The four shipped dictionary-update families as axis settings of the master problem (5). “Coupled” is the single-block consensus ($B=1$); “per-image” is $B=K$.

Family	partition (A)	engine (B)	inner (C)
ISM	coupled	consensus-ADMM	ism
CG	coupled	consensus-ADMM	cg
Cns	per-image	consensus-ADMM	sm
FISTA	coupled	proximal	—

- **Block partition** (axis A). The number of blocks B and their contents. $B = 1$ groups all images into one block, whose per-frequency Gram has rank K ; $B = K$ gives one image per block, of rank 1; an integer block size r (with $B = \lceil K/r \rceil$) gives blocks of rank r and interpolates between the two.
- **Enforcement engine** (axis B). *Consensus-ADMM* keeps f_b exact and moves $\iota_{C_{PN}}$ to the global variable, alternating per-block linear solves with a consensus averaging-plus-projection step. *Proximal* majorizes f by a Lipschitz quadratic and takes one proximal step of $\iota_{C_{PN}}$ on the primal variable (the FISTA/accelerated-proximal-gradient engine), with no linear solve at all.
- **Inner solver** (axis C , ADMM engine only). How each block’s diagonal-plus-low-rank system is solved: iterated Sherman–Morrison (*ism*, $O(r^2)$), conjugate gradient (*cg*, warm-started, tolerance 10^{-3}), or a single Sherman–Morrison step (*sm*, the $r=1$ case).
- **Basis** (axis D). The change of variables applied to the data term and constraint: *dft* (the standard spatial→DFT transform) or *block-dft* (an additional DFT along a coupling axis, which turns a block-circulant coupling into a block-diagonal one). The reviewed paper’s “3D” method is the block-dft value applied to the image axis.
- **Channel representation** (axis E , the multi-channel extension). Where the channel index lives: *single-channel-dictionary* (per-channel coefficient maps; the channel index folds onto the block index in the dictionary update) or *multi-channel-dictionary* (per-channel filters with one shared coefficient map; the update decouples over channels into independent grayscale updates sharing the same X).

One shared driver. The consensus-ADMM engine iterates, with over-relaxation α and penalty σ ,

$$\mathbf{d}_b \leftarrow \arg \min_{\mathbf{d}_b} f_b(\mathbf{d}_b) + \frac{\sigma}{2} \|\mathbf{d}_b - \mathbf{g} + \mathbf{h}_b\|_2^2, \quad (6)$$

$$\mathbf{g} \leftarrow \text{prox}_{\iota_{C_{PN}}/\sigma} \left(\frac{1}{B} \sum_b (\alpha \mathbf{d}_b + (1 - \alpha) \mathbf{g} + \mathbf{h}_b) \right) = P_{C_{PN}} \left(\frac{1}{B} \sum_b (\alpha \mathbf{d}_b + (1 - \alpha) \mathbf{g} + \mathbf{h}_b) \right), \quad (7)$$

$$\mathbf{h}_b \leftarrow \mathbf{h}_b + \alpha \mathbf{d}_b + (1 - \alpha) \mathbf{g} - \mathbf{g}, \quad (8)$$

where $\mathbf{h}_b$ is the scaled dual variable and $P_{C_{PN}}$ is the support-plus-normalization projection. The x-step (6) is solved in the DFT domain, where per frequency it is a diagonal-plus-rank- r_b system (r_b the number of images in block b). The proximal engine iterates

$$\mathbf{y} \leftarrow \mathbf{d} - \frac{1}{L} \nabla f(\mathbf{d}), \quad \mathbf{g} \leftarrow P_{C_{PN}}(\mathbf{y}), \quad (9)$$

with Nesterov momentum, the gradient computed in the DFT domain as $\nabla_{\hat{\mathbf{d}}} f = \hat{X}^H (\hat{X} \hat{\mathbf{d}} - \hat{\mathbf{s}})$. Both engines share the same f , the same $\iota_{C_{PN}}$, and the same $P_{C_{PN}}$; they differ only in whether the surrogate for f is a gradient majorizer or the exact quadratic plus an augmented-Lagrangian coupling. A single driver in the accompanying code (`cdl_template.py`) runs either engine from the axis values.

4 Recovering the known updates

The four update families shipped in the reference implementation are axis settings of (5) (Table 1).

Table 2: Primary results. For each recovered update and each representation: the axis settings that instantiate it, the recovery deviation against the reference (relative CDL-functional trajectory deviation and relative Frobenius error of the converged dictionary), and the converged functional value. The partial-consensus row reports the degenerate-collapse check (block size $r=1$ against the consensus reference and $r=K$ against the ISM reference) in place of an external reference.

Method	Repr.	Axis settings	Traj. dev.	Dict. err.	Conv. func.
ISM	gray	coupled, ADMM, ism	8.0×10^{-8}	3.5×10^{-7}	23.607
CG	gray	coupled, ADMM, cg	1.0×10^{-7}	3.4×10^{-7}	23.605
Cns	gray	per-image, ADMM, sm	1.1×10^{-7}	3.9×10^{-7}	24.078
FISTA	gray	coupled, proximal	7.8×10^{-8}	8.5×10^{-7}	24.441
Multi-dict	color	repr. = multi-channel-dict, coupled, ism	1.4×10^{-7}	8.7×10^{-7}	163.42
Single-dict	color	repr. = single-channel-dict, coupled, ism	1.1×10^{-7}	2.2×10^{-7}	213.31
Partial cns. ($r=2$)	gray	partition $r=2$, ADMM, ism (collapse)	$4.9 \times 10^{-7}/3.3 \times 10^{-7}$		32.41

The equality-constrained ADMM update and the consensus update are the $B=1$ and $B=K$ cases of (5). With $B=1$ the constraint $\mathbf{d}_1 = \mathbf{g}$ is exactly the equality constraint of the equality-constrained derivation, and the x-step (6) is the full rank- K solve; with $B=K$ the x-step splits into K independent rank-1 solves (Sherman–Morrison) and the y-step (7) is the consensus averaging-plus-projection. The two “different” ADMM derivations are one derivation read at two points of the partition axis. The FISTA update is the same master problem with the proximal engine on the coupled partition.

Validation. We instantiate the template through one shared driver and compare, on identical inputs (same seed, initial dictionary, images, λ , penalty/step, iteration budget, and inner-solver configuration), the CDL functional trajectory and the converged dictionary against the reference solver [1]. The sparse-coding stage is the same held-fixed solver in both runs, so the comparison isolates the dictionary update. On highpass-filtered MIRFLICKR-derived grayscale patches ($K=5$ images of 32×32 , $M=8$ filters of 8×8 , $\lambda=0.1$, 20 alternations, penalty $\sigma=K$ for the ADMM updates and inverse step $L=14K$ for FISTA, with the adaptive-penalty schedule disabled in both runs), all four families track the reference trajectory to relative deviation $\leq 10^{-7}$ and match the converged dictionary to Frobenius error $\leq 10^{-6}$ (Table 2, Figure 1). The exact DFT solves (ISM, consensus) agree to near machine precision on synthetic inputs and to $\sim 10^{-7}$ on the image data; the difference is floating-point conditioning of the data, not an algorithmic discrepancy.

5 Multi-channel instantiations

Real signals are multi-channel, and there are two genuinely different ways to add channel information. Both are instances of the same template, distinguished only by axis E .

Multi-channel dictionary with a shared coefficient map. The filters are $\mathbf{d}_{c,m}$ (channel c , filter m) and there is one coefficient map $\mathbf{x}_{k,m}$ per image shared across channels. The dictionary update is

$$\arg \min_{\{\mathbf{d}_c\}} \frac{1}{2} \sum_{k,c} \|X_k \mathbf{d}_c - \mathbf{s}_{k,c}\|_2^2 + \sum_c \iota_{C_{\text{PN}}}(\mathbf{d}_c), \quad (10)$$

which decouples over c into C independent grayscale updates sharing the same X_k (hence the same per-frequency factor), with joint channel normalization in the projection. In the template this is the coupled partition with the channel index carried on the filter axis; the per-channel solves share one rank- K Gram, so the update is no more expensive than grayscale.

Single-channel dictionary with per-channel coefficient maps. The filters $\mathbf{d}_m$ are single-channel and each channel has its own maps $\mathbf{x}_{c,k,m}$. The update is

$$\arg \min_{\mathbf{d}} \frac{1}{2} \sum_{c,k} \|X_{c,k} \mathbf{d} - \mathbf{s}_{c,k}\|_2^2, \quad (11)$$

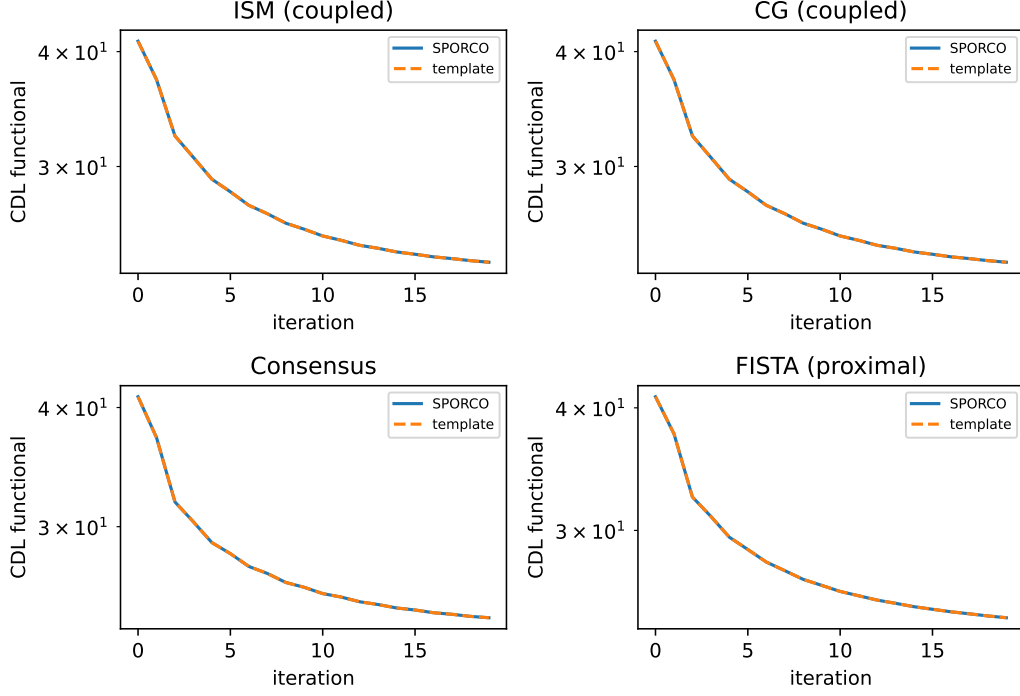


Figure 1: CDL-functional trajectories for the four recovered update families, each template instance overlaid on its reference run (identical inputs).

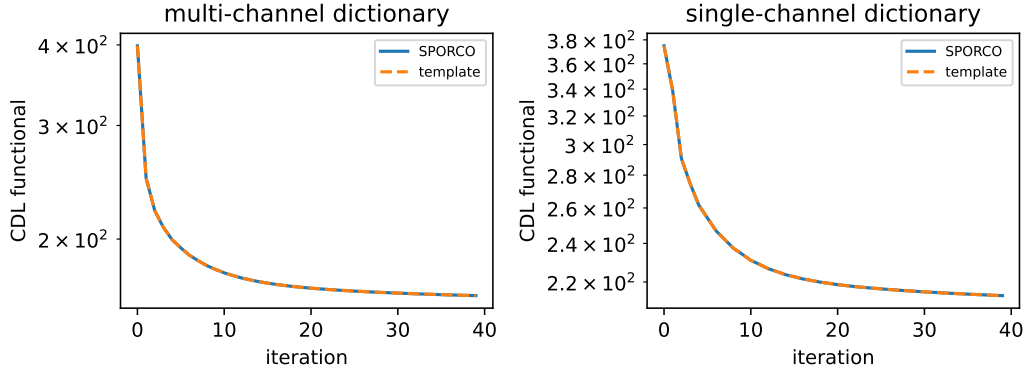


Figure 2: CDL-functional convergence for the two multi-channel representations, template overlaid on reference (color, $\lambda=0.1$).

where the channel index c now lives on the coefficient maps. The template folds c onto the block index, so the update is one grayscale update with CK data blocks and a rank- CK Gram. This is the point where the $K \leftrightarrow C$ duality is made explicit: the channel count enters the block axis exactly as the image count does.

Validation. On highpass-filtered color images ($C=3$, $K=4$ images of 64×64 , $M=8$ filters of 8×8 , $\lambda=0.1$), both representations are realized from the template and run to convergence: the multi-channel-dictionary run reduces the functional to $0.41 \times$ its initial value and matches the reference to Frobenius error 8.7×10^{-7} ; the single-channel-dictionary run reduces it to $0.57 \times$ and matches to 2.2×10^{-7} (Figure 2). The reference for the single-channel representation is a plain-CBPDN run (no joint-sparsity coupling), and both runs use the frozen $\lambda=0.1$.

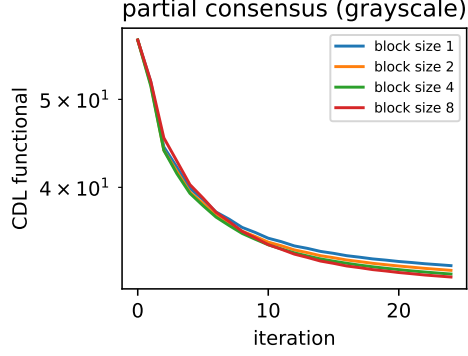


Figure 3: The partial-consensus cell: CDL-functional trajectories for block sizes $r \in \{1, 2, 4, 8\}$ on grayscale data. All run and converge; $r=1$ and $r=8$ collapse to the consensus and ISM references respectively.

6 What the template predicts

A template is worth more than the methods it was built from only if its empty cells generate working methods. The partition axis has an empty cell: a *partial* consensus with block size $1 < r < K$, whose blocks carry a rank- r Gram. No shipped or reviewed method occupies this cell (the shipped methods are $r=1$ and $r=K$).

The partial-consensus update. With block size r , the x-step solves $B = \lceil K/r \rceil$ independent diagonal-plus-rank- r systems by iterated Sherman–Morrison, and the y-step averages over B blocks. It runs and converges (Figure 3). Because the cell has no external reference, it is validated by the degenerate collapse the template requires: setting $r=1$ must reproduce the consensus reference and $r=K$ the ISM reference. Both hold to $\leq 10^{-6}$ (Table 2), which is the decisive check that the template generates the method rather than dispatching to a pre-written branch.

A re-derived equivalence. The template makes the reviewed paper’s central observation structural. The equality-constrained ADMM update and the consensus update are not merely “the same class”; they are the same master problem (5) at $B=1$ and $B=K$. This follows by reading (6)–(8) at the two endpoints: at $B=1$ the consensus averaging is trivial and the constraint $\mathbf{d}_1 = \mathbf{g}$ is the equality constraint of the equality-constrained derivation; at $B=K$ the blocks decouple into per-image rank-1 systems and the averaging is the consensus step. In the multi-channel setting the same reading applies with the channel index folded onto the block axis: the single-channel-dictionary update is the grayscale update with $K \rightarrow CK$, so the $B=1$ and $B=CK$ endpoints recover the multi-channel ISM and consensus, and the $B=K$ cell (block size C) is the intermediate. This is the $K \leftrightarrow C$ duality of the reviewed work, expressed as a consequence of axis E (where the channel index lives) rather than asserted.

7 Why the axes matter

The axes are not bookkeeping; they control cost and trajectory through the linear-system structure. The expensive quantity is the per-frequency rank r_b of each block’s Gram, summed over blocks. With block size r there are K/r blocks of rank r , so the iterated-Sherman–Morrison linear cost is $O(KrMN)$ per update— $O(KMN)$ at $r=1$ (consensus), $O(K^2MN)$ at $r=K$ (ISM)—while the number of consensus synchronization points is K/r . Conjugate gradient replaces the $O(r^2)$ solve by $O(r n_{cg})$ matvecs and is the right choice when r is large and warm-starting is effective; the proximal engine performs no solve at all, at the cost of a gradient step whose Lipschitz constant scales with the block count. All methods remain FFT-bound at $O(KMN \log N)$.

This cost structure explains the reviewed paper’s empirical ordering, and it is complemented by a finding that the separate derivations obscure: the dictionary-update subproblem is a convex quadratic over the unit sphere $\|\mathbf{d}_m\|_2 = 1$, hence non-convex in the constraint. Different splittings of a

non-convex problem follow different trajectories and can land at different stationary points. With the coefficient maps held fixed and the inner ADMM run to convergence, the block sizes $r \in \{1, 2, 4, 8\}$ reach dictionaries that differ by Frobenius distance $O(1)$ —not because the template is wrong, but because the block-size choice is a genuine algorithmic choice on a non-convex landscape. This is the mechanism behind why the choice of update algorithm matters at all, and it is precisely what a dispatcher over the known solvers would fail to reveal.

8 Related work

The structure we abstract is the one introduced and refined in [1] and its antecedents: the augmented-Lagrangian dictionary update with frequency-domain solves [2], the consensus formulation [3], and the accelerated proximal-gradient update [4]. The reviewed work already carries these updates to C channels (its Sec. VI), derives the multi-channel-dictionary representation, notes the $K \leftrightarrow C$ duality, and reports color experiments; that material is our baseline, not our contribution. Our contribution is the unification itself: one master problem whose axes generate the methods, the two representations, and—crucially—a method that none of the known algorithms or their multi-channel extensions occupy. The template’s partial-consensus cell and the re-derived equivalence are not present in the reviewed work or its follow-ups; what distinguishes this work is that the new method arises from an empty cell of a shared derivation rather than from a separate algorithm. We therefore position the template as the contribution and the recovery, multi-channel runs, and collapse check as the verifiable anchors.

The operator-splitting and majorization-minimization frameworks we use are classical [5, 6]; our use of them here is only to make explicit a design space that the CDL literature has explored one algorithm at a time.

9 Discussion and limitations

The template is exact for the dictionary update, which is the contribution’s object; it does not abstract the sparse-coding stage or the alternation strategy, which are held fixed. The inter-channel-correlation representation (per-channel maps coupled by a cross-channel ℓ_1 term) is not instantiated: its distinguishing content lives in the sparse-coding subproblem, which does not exercise the dictionary-update template. The “3D” (block-dft) method is recovered only as the basis-axis equivalence that the reviewed paper already established; we do not re-implement it as a new cell because the channel axis of the single-channel-dictionary update has no block-circulant structure to diagonalize—the fold is a re-indexing, not a block DFT, and we state this rather than over-claim a channel-“3D” method. The adaptive-penalty schedule of the reference was fixed (identical in both runs) so the comparison isolates the update iteration; this is disclosed. All quantitative claims trace to the logged runs in the accompanying repository.

10 Conclusion

We have shown that the differently-derived algorithms for the convolutional dictionary update are instances of one operator-splitting master problem, and we have made that structure explicit as a template parameterized by a handful of design axes. The template reproduces the shipped updates to near machine precision through one shared driver, expresses both multi-channel representations of the problem, and predicts a partial-consensus update that no known method occupies, validated by its degenerate collapse to the two reference endpoints. The equivalence it re-derives—equality-constrained and consensus updates as the two ends of one block-partition axis, with the channel count entering that axis exactly as the image count does—is the reviewed paper’s central observation, made structural rather than incidental.

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